

BINOMIAL $\rightarrow$ NORMAL

(for large n and npq not too small)

$$P(X=k) = \binom{n}{k} p^k q^{n-k}$$

$X \sim \text{binomial}(n, p)$

successes in n indep. Bernoulli trials

$X = B_1 + B_2 + \dots + B_n$ where $B_i \sim \text{Bernoulli}(p)$

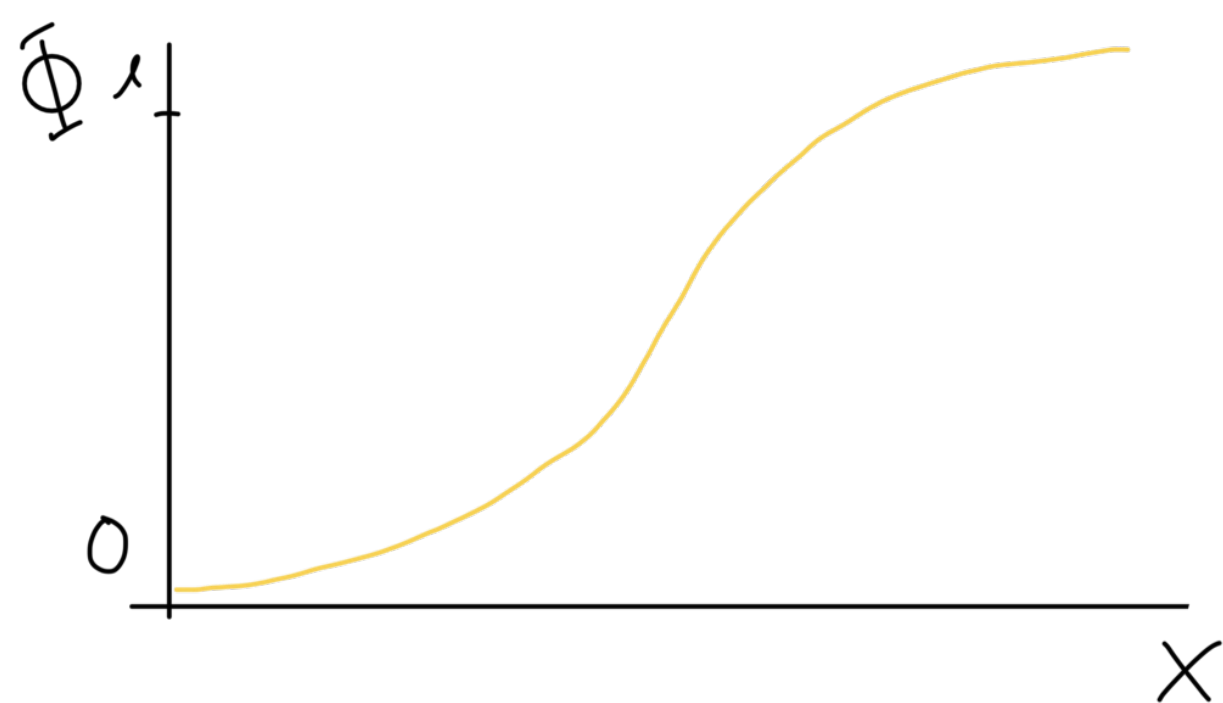
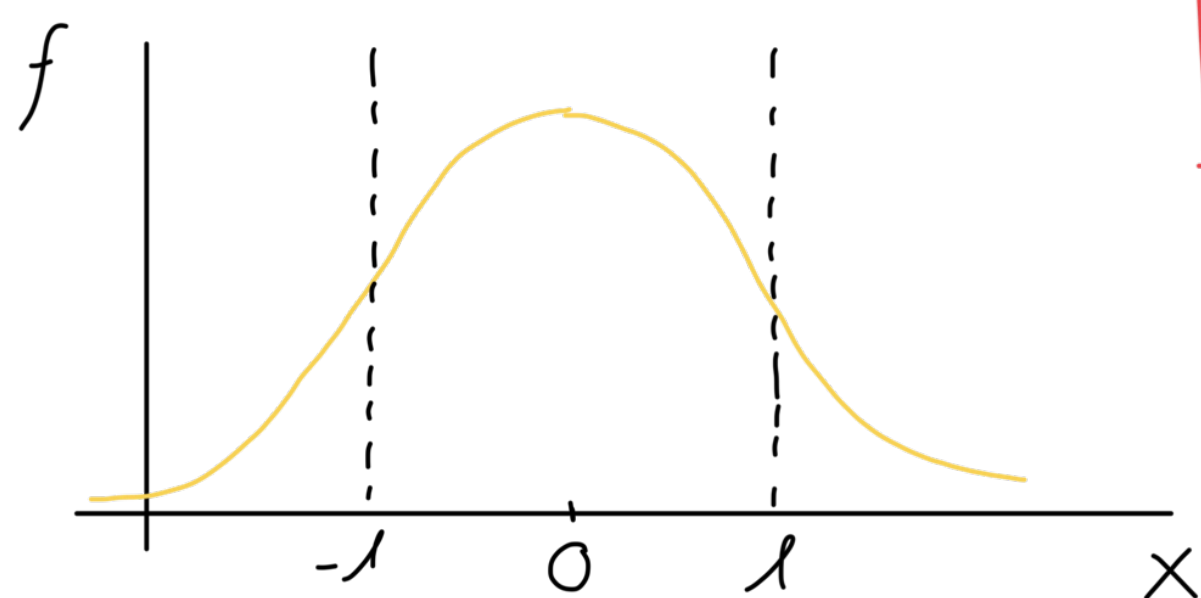
$X \sim \text{normal}(np, npq)$

μ σ^2

$$P(X=x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

EASIER TO COMPUTE
WITH $N(\mu, \sigma^2)$!

STANDARD UNIT NORMAL ($\mu=0, \sigma=1$)



$$P(X \leq x) = \int_{-\infty}^x f(x) dx$$

$$P(-1 \leq X \leq 1) = \int_{-1}^1 f(x) dx = \Phi(1) - \Phi(-1)$$

Ex:

FLIP A COIN 400 TIMES ; # HEADS = $X \sim \text{binomial}(400; 1/2)$

$$P(190 \leq X \leq 230) = P(X=190) + P(X=191) + \dots + P(X=230)$$

$X \sim \text{normal}(200, 100)$

